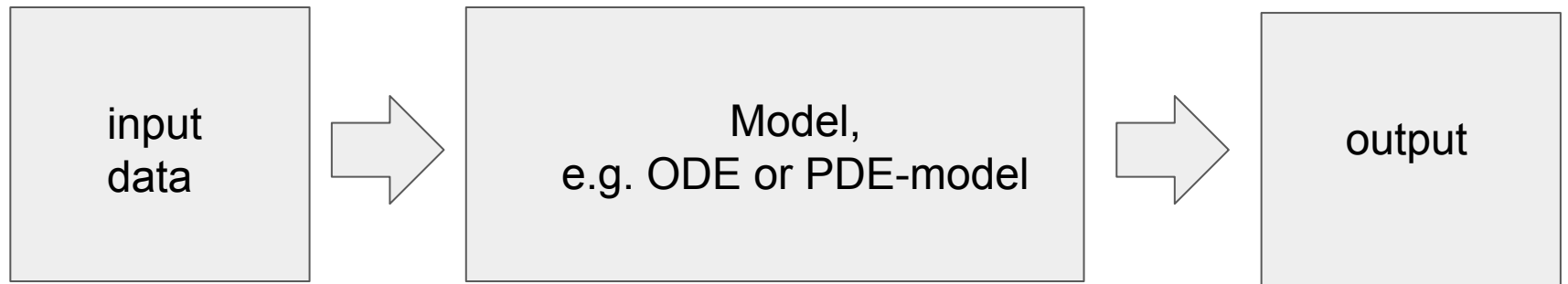


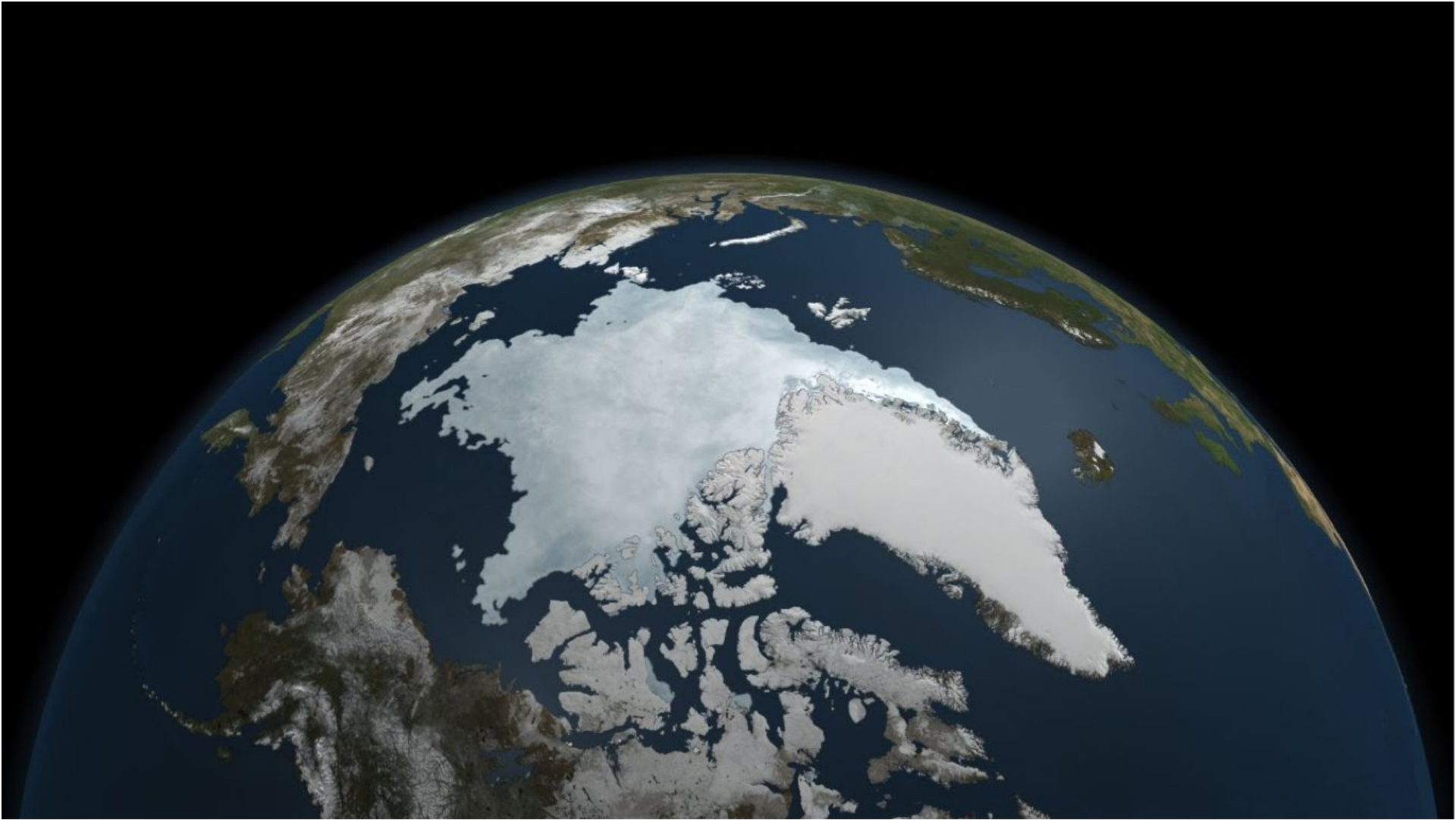
Inverse modelling

Application to Ice Sheet Modelling

Forward models



Ice sheet modelling



<https://nsidc.org/data>

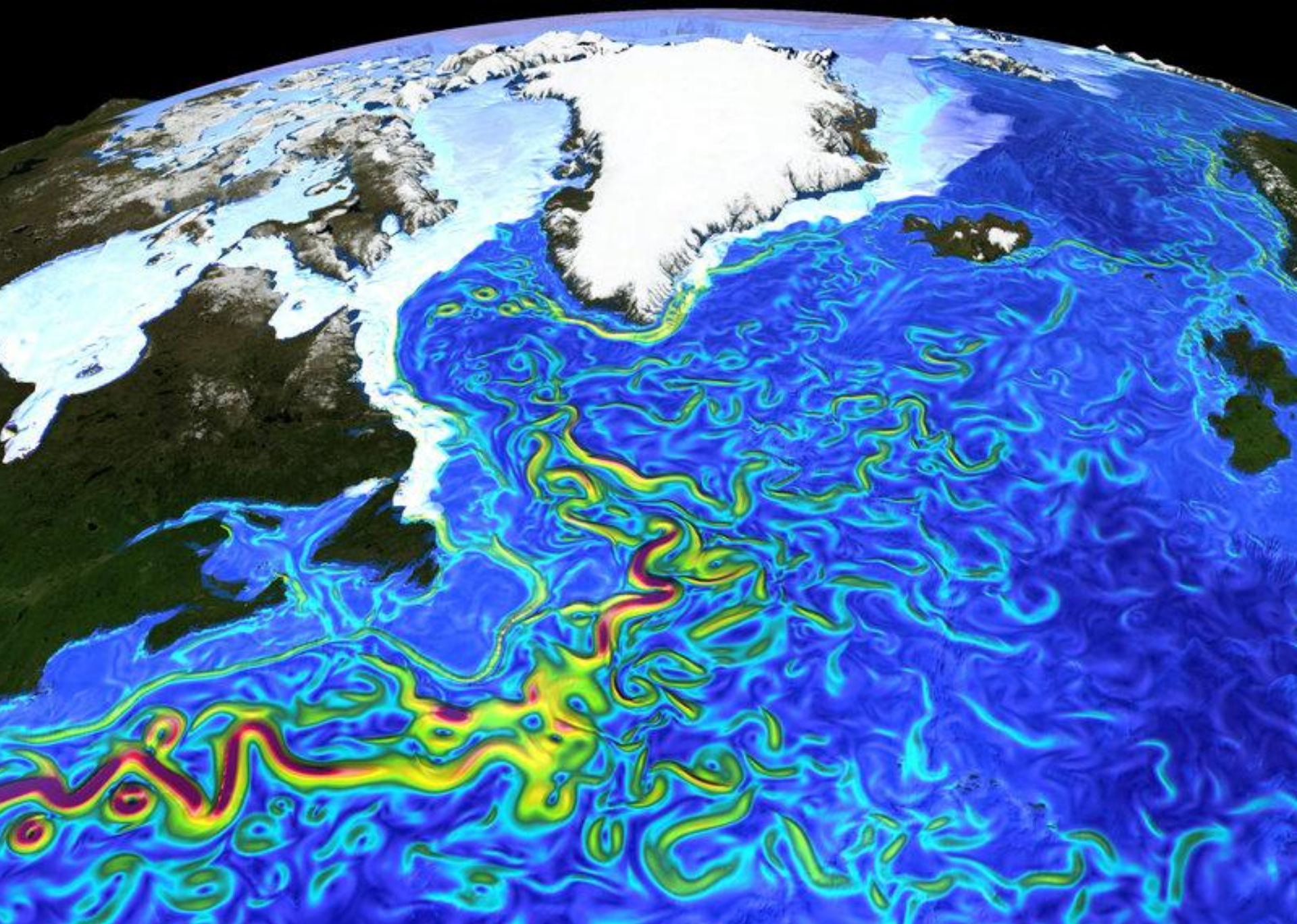
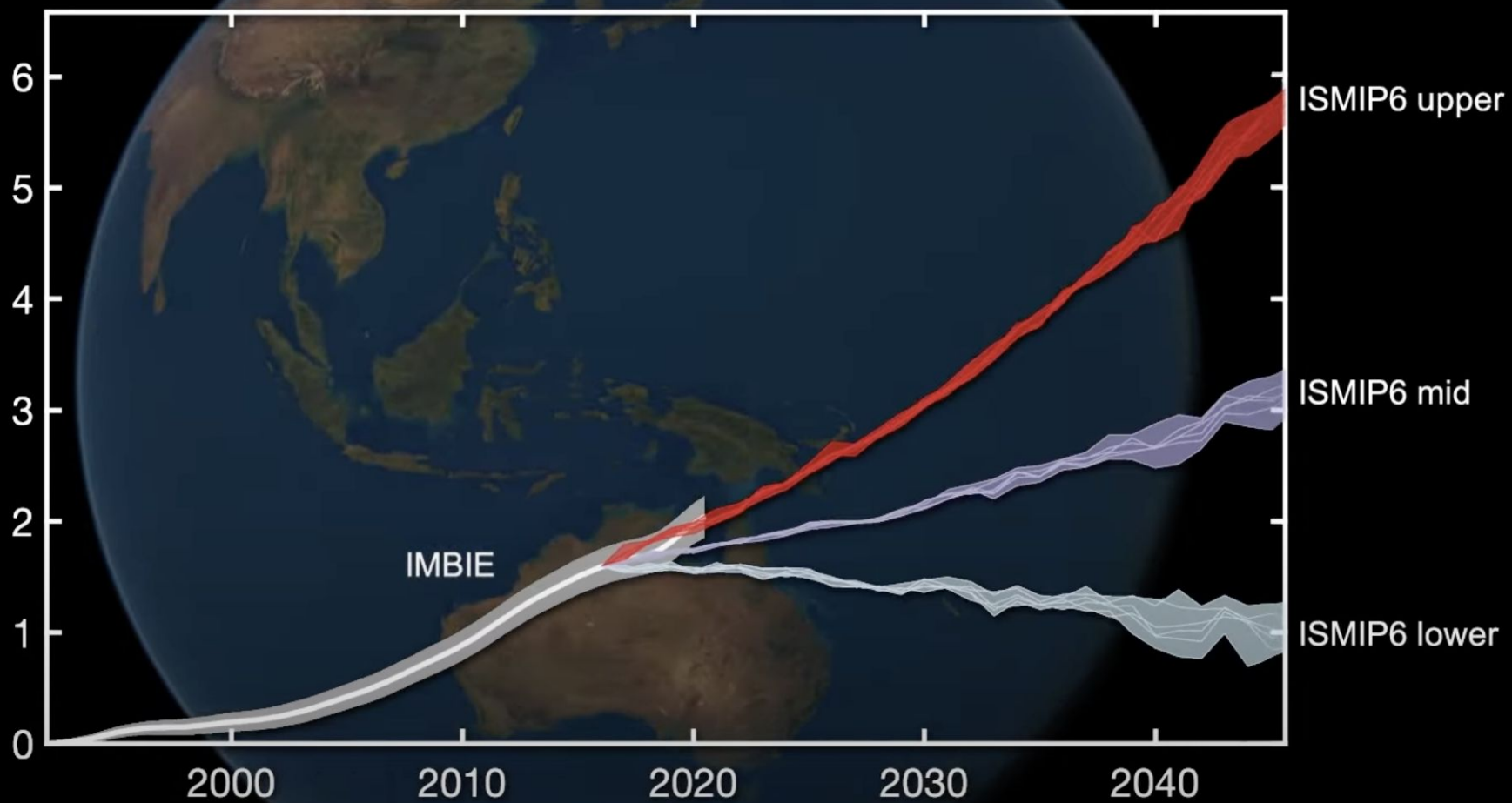
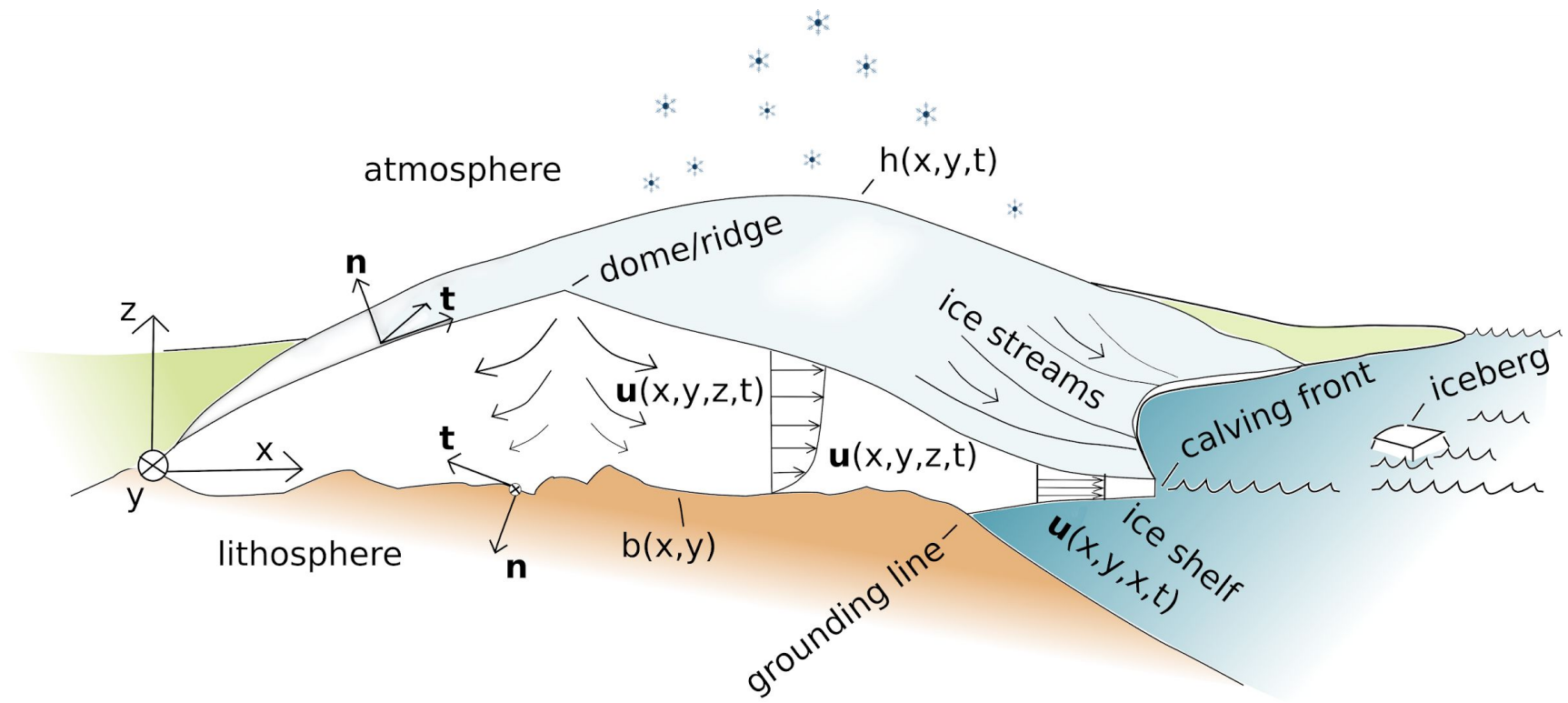


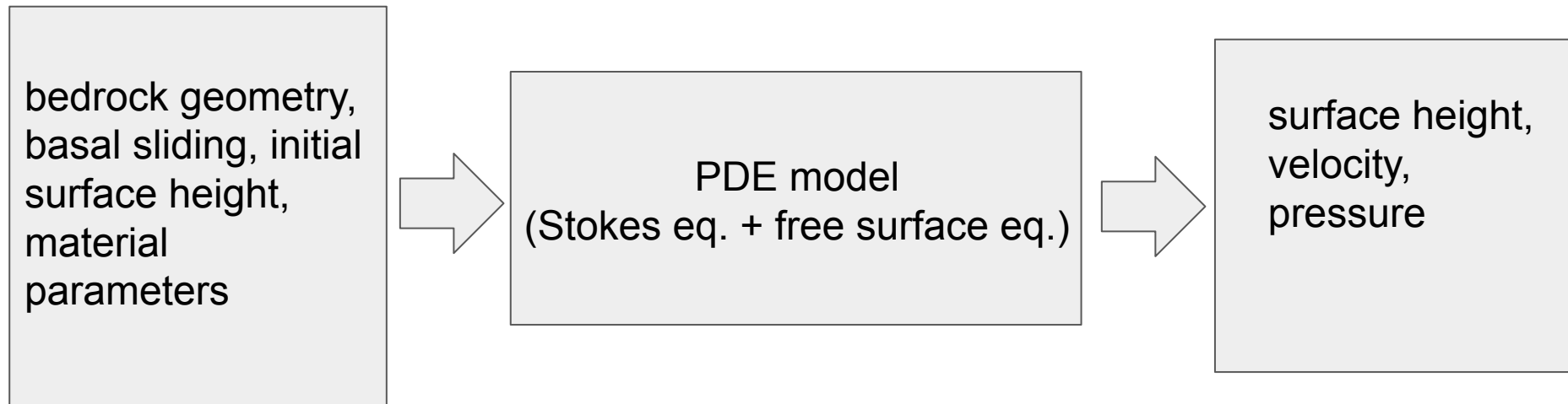
Bild från: Böning, C. W., E. Behrens, A. Biastoch, K. Getzlaff, J. L. Bamber (2016): Emerging impact of Greenland meltwater on deepwater formation in the North Atlantic Ocean. *Nature Geoscience*, [dx.doi.org/10.1038/ngeo2740](https://doi.org/10.1038/ngeo2740)

Ice Sheets Contribution to Global Sea Level





Forward model, ice



$$\frac{du}{dx} + \frac{d^2u}{dx^2} = f$$

ODE

$$\frac{\partial u}{\partial x} + \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial t} = f$$

PDE

Forward model: the p-Stokes equations+free surface eq.

The diagram shows the p-Stokes equations with several annotations. The first equation, $-\nabla \cdot \mathbf{S}(\mathbf{Du}) + \nabla p = \rho \mathbf{g}$, is annotated with 'pressure gradient' pointing to ∇p , 'gravity' pointing to $\rho \mathbf{g}$, and 'non-linear stress tensor' pointing to $\mathbf{S}(\mathbf{Du})$. The second equation, $\nabla \cdot \mathbf{u} = 0$, is annotated with 'divergence free velocity' pointing to $\nabla \cdot \mathbf{u}$. Both equations are defined 'in $\Omega(h(t))$ '.

$$\begin{aligned} -\nabla \cdot \mathbf{S}(\mathbf{Du}) + \nabla p &= \rho \mathbf{g} && \text{in } \Omega(h(t)) \\ \nabla \cdot \mathbf{u} &= 0 && \text{in } \Omega(h(t)) \end{aligned}$$

$$\frac{\partial h}{\partial t} + u_x(h) \frac{\partial h}{\partial x} + u_y(h) \frac{\partial h}{\partial y} = u_z(h) + a_s$$

The Shallow Ice Approximation (SIA)

$$v_x = v_{b,x} - 2(\rho g)^n \frac{\partial h}{\partial x} \|\nabla h\|^{n-1} \int_b^z \mathcal{A}(T')(h - z')^n dz',$$

$$v_y = v_{b,y} - 2(\rho g)^n \frac{\partial h}{\partial y} \|\nabla h\|^{n-1} \int_b^z \mathcal{A}(T')(h - z')^n dz',$$

$$v_z = v_{b,z} - \int_b^z \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} \right) dz',$$

$$p = \rho g(h - z).$$

plug into the free surface equation as before

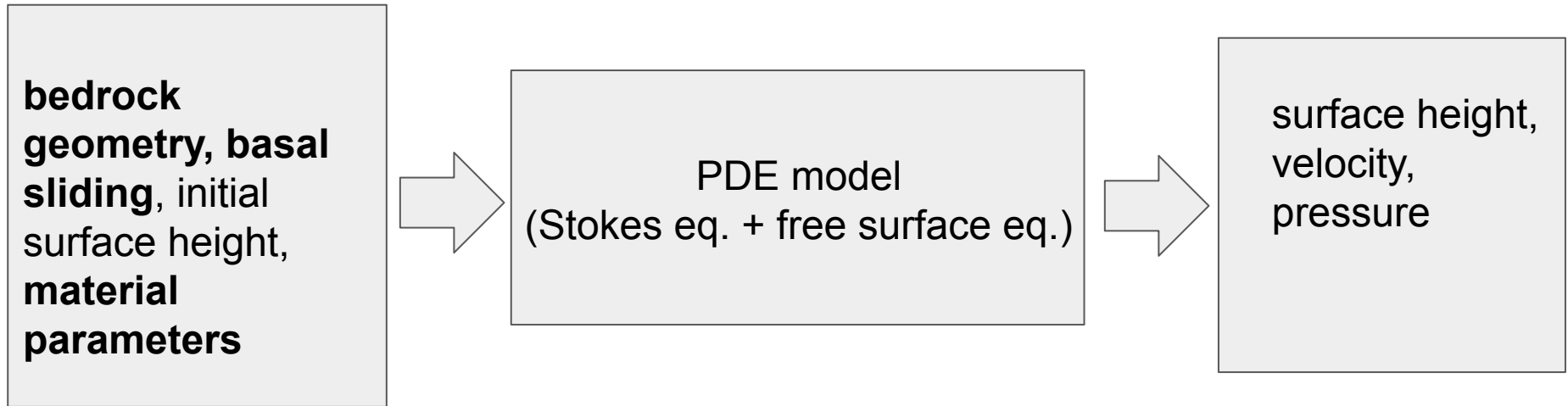
Code test:

Download

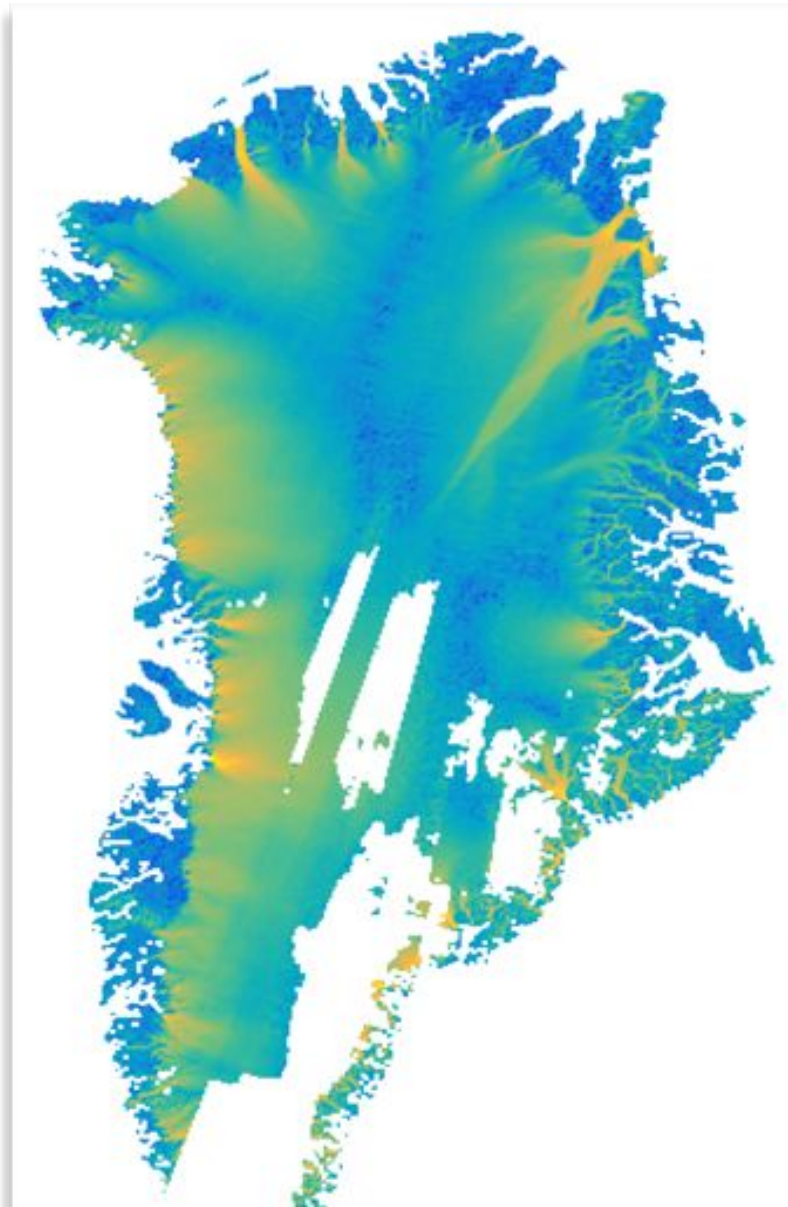
- RunIceSheetSimulation.py
- icemodel.py

from the course webpage

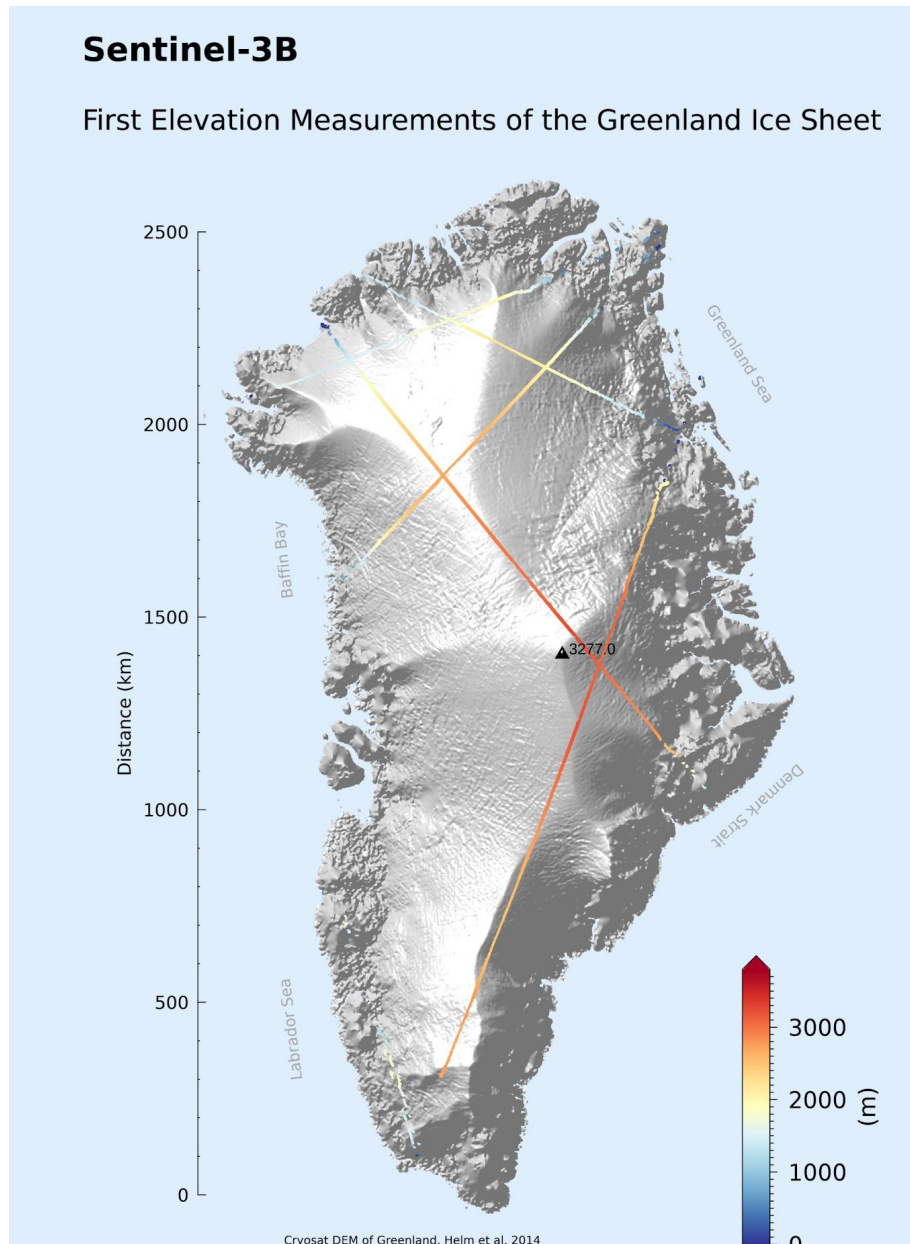
Problem - some input data is hard to collect



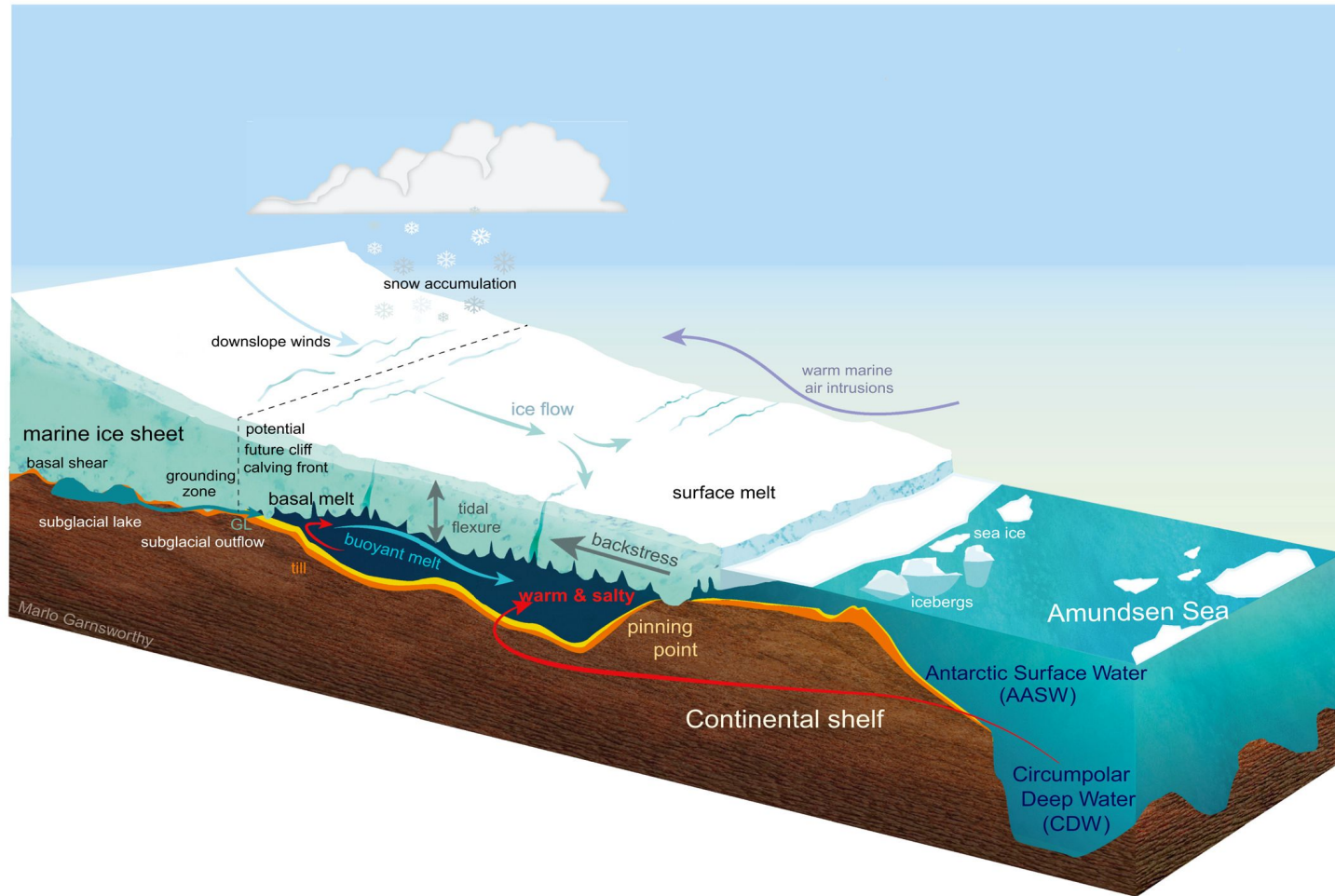
Satellite observations of ice velocity at the surface

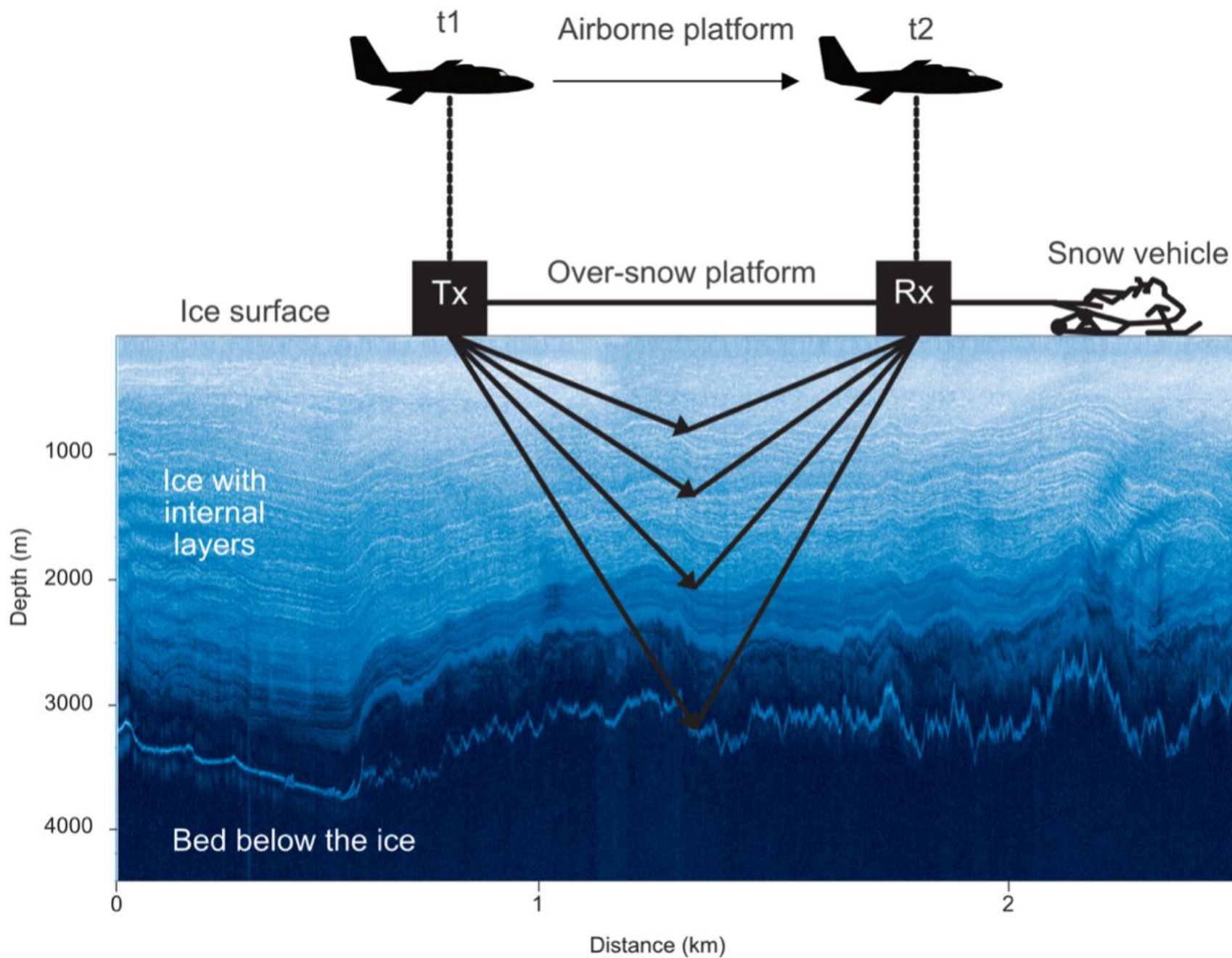


Satellite observations of surface position

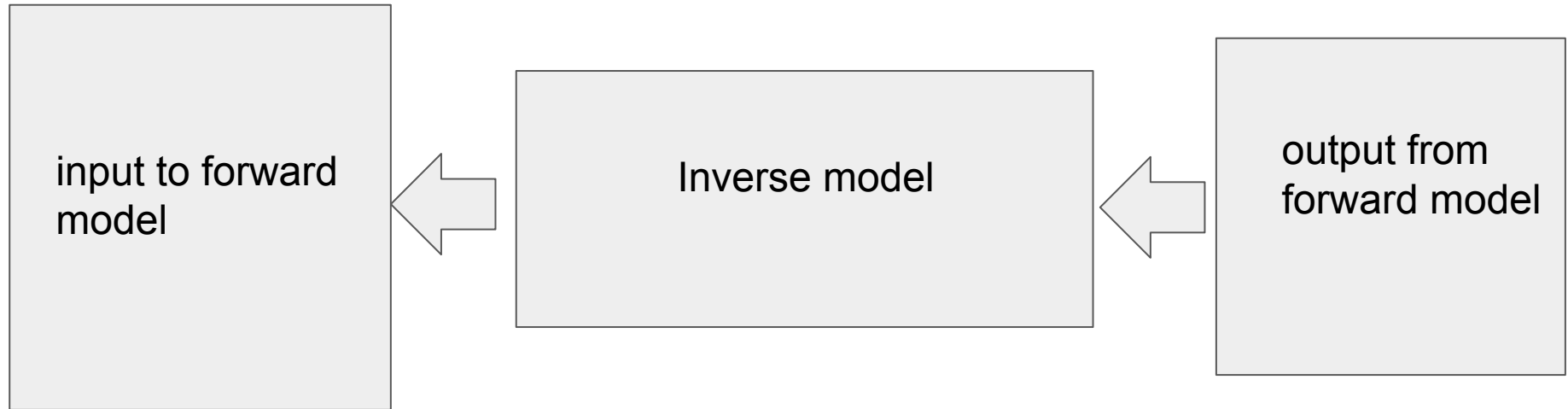


Bed properties hard to observe

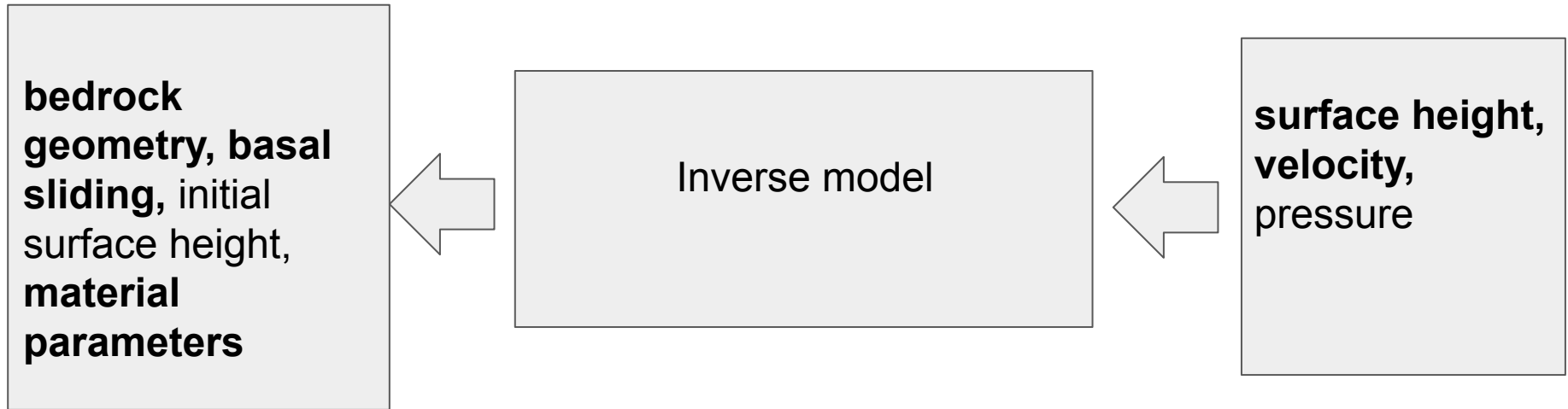




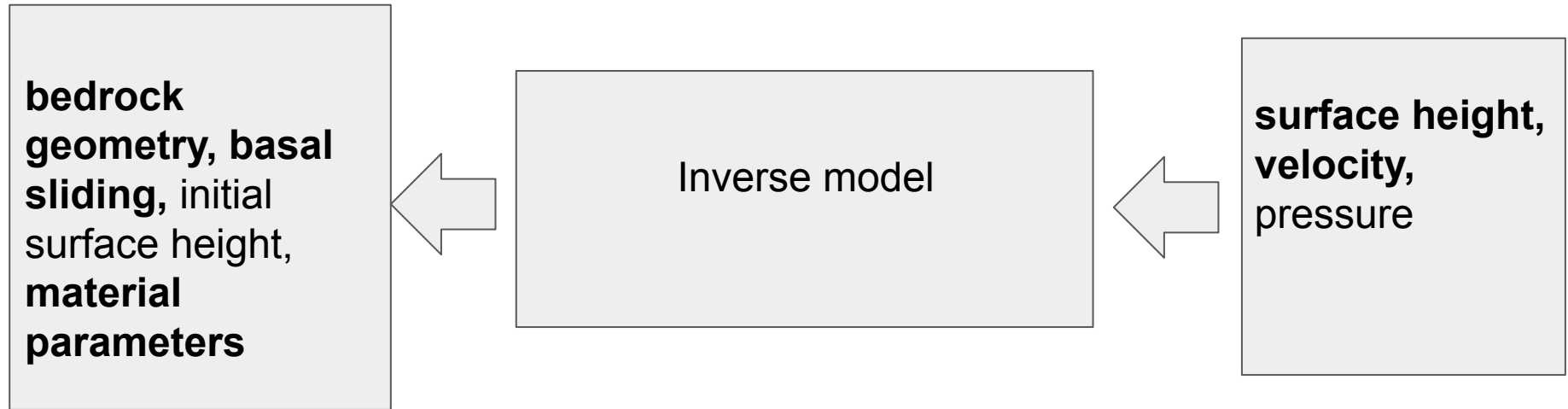
Solution: “inverse modelling”



Inverse model, ice



Inverse model, ice



To think about - is the solution unique?

Task - part 1

Create the inverse model!

- Download data for surface position and surface velocity from the course webpage. Determine what the bedrock shape must have been.
- Assume the bedrock is created as in `icemodel.CreateBedrock`, and that the only uncertain parameters are the amplitudes of the bumps
- Do this however you like, **but you must be able to explain your reasoning in the oral presentation**. Next time we'll talk a bit about fancy ways.
- Hint: I did not tell you at what year this data was retrieved - i.e. ***you should not consider time evolution*** (this would be computationally costly).
- To test your model, you can create your own data with the forward model (uncomment the lines for saving data).

Task - part 2

- In reality, the data contains measurement errors
- Add some random noise (a few percent) to the data - how does that impact your ability to determine the bedrock amplitudes?

Hand in

- Your answer, i.e. the bedrock amplitudes
- A plot or table which explains why you chose those amplitudes
- A plot with the forward model for your chosen bedrock
- A brief explanation of your method
- Your code
- Some plot or table which describes how noise impacts your inverse model, together with a few sentences.

Next time

- Oral presentations: 10 minutes per group
 - Explain your method clearly
 - Explain the structure of your code
- A small lecture on fancy inverse models

Code walk-through